

Material Specification — Inconel 718 Bar (Incoming)

Incoming material requirements and per-lot certification rules for Inconel 718. Approved supplier: SUP-AERO-METALS .

```
--- METADATA ---
document_id: ERP-MAT-INC718-SPEC
title: Material Specification — Inconel 718 Bar (Incoming)
domain: materials
source_system: ERP
revision: Rev C
effective_date: 2024-12-01
owner: K. Nakamura, Materials Engineering — Wichita
part_numbers: [45621-B]
operation_numbers: []
machine_ids: []
controller_models: []
fault_codes: []
materials: [MAT-INC718]
standards: [AS9100D]
supplier_ids: [SUP-AERO-METALS]
jurisdictions: []
classification: internal
---
```

1. Specification reference

Field	Value
UNS designation	N07718
AMS specification	AMS 5662 (bar, solution & precipitation heat-treated)
Allowed bar diameters	Ø 50, Ø 75, Ø 100 mm
Allowed bar length	2.0 m and 3.0 m
Heat-treat condition on receipt	Solution treated & aged (STA), 36–46 HRC

2. Per-lot certification requirements (mandatory)

Every production lot must arrive with the following documentation. Material may not be released to production until all three are on file (AS9100D §8.5.6 traceability + Vortex / Aeronova CSR §5.4 for safety-critical parts).

- 1. **Mill test report (MTR)** — chemistry, mechanical properties (UTS, YS, %EL), heat-treat record. Must include heat number and lot number.
- 2. **Nadcap heat-treat certification** — supplier's Nadcap accreditation current for the heat-treat process used.
- 3. **Conflict-minerals declaration** — for nickel-content alloys, supplier statement per Vortex Motors VSQM §5.4.

For the *next production batch* of Part 45621-B (which consumes this material), the corrective-action process described in [QMS-IATF-EXCERPT-001](#) §10.2 requires that the per-lot certification above be verified before the batch is released, particularly when the previous batch had any open NCR.

3. Incoming inspection

- 1. Visual: surface defects, oxide pockets, bar straightness.
- 2. Dimensional: diameter (± 0.5 mm), length (+10 / -0 mm).
- 3. Hardness: 5 points per bar; 36–46 HRC.
- 4. PMI (positive material ID): handheld XRF on first bar of each heat.

4. Storage

Vertical rack, segregated by heat number. Bars must remain identified with supplier-applied colour tag until consumed. Mixed-lot storage is not permitted.

Document control / sign-off

Role	Name	Signature	Date
Document Owner	K. Nakamura, Materials Engineering — Wichita	(on file)	2024-12-01
Approver	K. Nakamura, Materials Engineering — Wichita	(on file)	2024-12-01